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Sophie Ebersold (IRIT)
Jannik Laval (DISP Lab)

Subject: Application for PhD Position – Integration and synchronization of digital twins for co-simulation

Dear Sophie Ebersold and Jannik Laval,

I am applying for the PhD position on "Integration and synchronization of digital twins for co-simulation" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

The orchestration of disparate digital twins into a cohesive co-simulation is a challenge I am eager to solve. Having engineered systems that federate raw telemetry, Graph Neural Networks, and thermodynamic models, I recognize the need for robust semantic synchronization. I look forward to developing formal synchronization protocols that allow independent DTs to interact within the Artemis platform without losing temporal or semantic fidelity.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui